Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Smart Fiber Optical Sensor**: Internal Fuel Cell Multiparameter Sensor system for state-of-operation monitoring and state-of-health diagnosis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Energy & REsources	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Optical Measuring Instruments	New & Renewable Energy	Fuel Cells
Concept of Proposed Technology	TNO's optical fiber based multiparameter sensor (OFMS) platform enables **affordable, scalable, flexible multi-point state of operation (SOO) monitoring**—far beyond traditional electrical single-point sensors, which are hard to scale and risky in hydrogen environments. It provides multi-parameter readouts such as **temperature, pressure, strain** and **hydrogen concentration** in a single system. Due to the small size and inert material property (silica), optical fiber can be integrated in hydrogen fuel cell to achieve and obtain direct real-time monitoring of critical operational data from inside of a fuel cell. Enabled by OFMS, flexible and reactive operation control become possible for improved efficiency. The time logged monitoring data over sufficient time will form a solid base as diagnostic system for **state-of-health (SOH)** analysis. With accurate and forecasting information, preventive maintenance can be realised.			

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Rationale for Joint R&D Project	Hydrogen fuel cell monitoring is moving toward multi-parameter, intrinsically safe, scalable, and high-performance sensing—exactly the capabilities embedded in TNO's optical fiber technologies. South Korea, meanwhile, has one of the world's most active hydrogen technology supply chains, centered around agile SMEs in materials, sensors, fuel cell components, and system integration. South Korea is among the world's most advanced hydrogen economies, backed by strong government commitment, rapid technology adoption, and major industrial participation. Integration of IoT, AI, predictive maintenance into fuel cell engines is accelerating adoption and improving cost-efficiency. However, state-of-the-art fuel cell operation monitoring is limited by indirect measurement methods with delay between measurement taken and information delivery. The TNO OFMS system has reached TRL 4 and has been successfully tested in research-grade fuel cell systems. It is ready to be explored and implemented to large industrial testing for targeted user specification and requirements. South Korea's fuel cell and hydrogen industry, including SMEs, can leverage TNO expertise while offering diverse test cases through collaboration. TNO holds key IPs on OFMS system including: patent [1], dedicated tools and engineering methods, most advanced research and testing labs and facilities and more than 40 years Know-Hows in Fiber Optics domain.
Key Words	1. Optical fiber sensor 2. Fuel cell 3. State-of-operation monitoring 4. State-of-health diagnostics 5. Photonics technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	Hydrogen fuel cells, particularly proton exchange membrane fuel cells (PEMFCs), are widely regarded as a cornerstone technology for low-carbon transportation and stationary power generation. Despite substantial advances in materials and system design, durability and reliability under real-world operating conditions remain critical barriers to large-scale deployment. Consequently, accurate state-of-operation (SoO) monitoring and state-of-health (SoH) diagnostics have become central research topics in fuel cell systems engineering.[2, 3] SoO monitoring concerns the real-time observation of operating variables such as voltage, current density, temperature, pressure, and reactant stoichiometry. Although these variables are generally measurable, their diagnostic value is limited by strong nonlinear coupling between electrochemical, thermal, and water management processes. Similar SoO signatures can arise from distinct phenomena such as flooding, dehydration, hydrogen starvation, or transient load changes, making fault isolation inherently ambiguous [2]. This ambiguity is exacerbated in automotive applications, where highly dynamic drive cycles introduce reversible effects—such as temporary voltage recovery—that mask irreversible degradation [4]. Furthermore, sensor limitations, including drift, noise, and restricted accessibility to internal states (e.g. membrane hydration or local oxygen transport resistance), constrain the fidelity of SoO-based diagnostics [2, 3] SoH diagnostics pose greater challenges, as fuel cell degradation mechanisms are internal, cumulative, and spatially distributed. Key aging processes—including membrane chemical degradation, catalyst dissolution, and carbon corrosion—cannot be directly observed during normal operation and often produce overlapping macroscopic effects [4]. Unlike batteries, fuel cells lack a standardized, universally accepted SoH metric. Existing proxies such as voltage decay rate, ohmic resistance increase, or electrochemical surface area (ECSA) loss are condition-dependent and require intrusive or quasi-steady tests, limiting their applicability for online monitoring [2, 3]. As a result, SoH estimation is fundamentally an inverse and ill-posed problem. The challenges of state-of-operation (SoO) and state-of-health (SoH) diagnostics in PEM fuel cells translate primarily into stringent sensor requirements for internal observability, environmental robustness, and long-term stability. SoH degradation originates from localized, microscopic phenomena that cannot be directly monitored with conventional electrical sensors, while dynamic operating conditions introduce strong cross-sensitivities among temperature, humidity, and load, complicating signal interpretation. Consequently, sensor limitations—rather than algorithms alone—often dominate diagnostic uncertainty. Within this context, optical fiber sensors (OFS) offer distinct advantages. Their electrical passivity, immunity to electromagnetic interference, and resistance to humid and chemically aggressive environments make them well suited for in-situ deployment in fuel cell stacks. Distributed fiber sensing and fiber Bragg gratings enable spatially resolved measurements of temperature, strain and other parameters, improving detection of local hot spots and mechanical stresses linked to degradation. In combination with electrochemical sensors, the optical fibre multiparameter sensors (OFMS) enabled multimodal sensor system can help to improve the SoH diagnostic quality. This project aims to develop and integrate OPMS into fuel cells for SOO monitoring and SOH diagnostics. The OPMS can be further integrated in AI enabled Multimodal sensors system to achieve an improved accuracy and coverage of the operation. While the specific application in the hydrogen fuel cell (FC) is targeted within this proposal, the innovative approach developed in this project is widely applicable, enabling a broad range of electrochemical processes and equipment.			
R&D Target Specifications	A Fiber Bragg Grating (FBG) [5] is a periodic modulation of the refractive index inscribed within the core of an optical fiber. This structure reflects a narrow spectral band of light—called the Bragg wavelength—while transmitting the other wavelengths. The Bragg wavelength (λ_B) shifts predictably as a function of strain and temperature, according to: $$\Delta\lambda_B = \lambda_B[(1-p_e)\varepsilon + (\alpha+\xi)\Delta T]$$			

where ε is strain, ΔT is temperature change, and the coefficients represent photo-elastic and thermo-optic effects. The wavelength response of FBGs is high stable, immune to electromagnetic interference, and suitable for harsh environment.

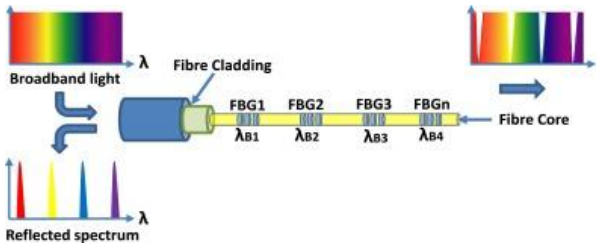

Figure 1 Multipoint FBG sensor array functional principle.

In this project, we aim to move beyond today's non-modular, single-parameter (and mostly external) systems and develop both multimodal (integrating different, complementary sensor types) and multiparameter (where one sensor system? can capture several physical parameters, directly or indirectly) sensing. Optical Fibre Sensors (OFS) are ideally suited to this transition, as they inherently enable compact, multiplexed, and versatile sensing. However, State-of-the-Art (SotA) OFS for in-cell measurement is limited to a research demonstration of temperature sensing. No commercial system is available. Building on our proof-of-concept demonstrations of in-cell, real-time OFS measurements in FC, we will go beyond the SotA to develop and integrate innovative OFMS solutions, including specialty fibres and advanced coating technologies. This will allow us to introduce cutting-edge systems expanding the dimensionality of sensing (multimodal sensing), improving data quality, and achieving significantly more accurate operational monitoring at lower system cost.

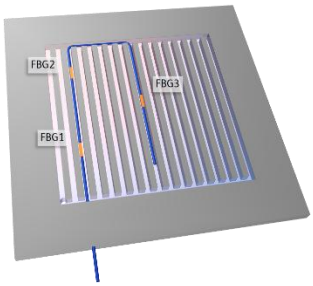

Figure 2 Embedded OFMS in fuel cell flow field channels.

Fuel cells and especially PEM fuel cells present an environment where conventional electrical sensors face limitations due to:

- High humidity and acidic conditions,
- Hydrogen and oxygen exposure,
- Electromagnetic noise from power electronics,
- The need for internal or near MEA sensing.

FBGs are electrically passive, chemically robust (silica-based), and can be embedded or surface-mounted without electrical connections inside the stack, making them particularly suitable for in-situ sensing.

FBG sensors measure operational parameters those are strongly correlated with FC performance and degradation. Through discrete FBG points in a single optical fiber, a multipoint sensor array of such can be used to measure **temperature and strain/mechanical stress** distribution inside FC. The temperature measurement can yield insights on, for instance, 1) detection of local hotspots in flow fields or bipolar plates; 2) identification of uneven reaction distribution and cooling efficiency; 3) persistent temperature gradients linked to accelerated membrane and catalyst degradation. The strain/mechanical stress has correlation to other physical effects, for instance, 1) compression changes due to stack assembly and swelling; 2) Membrane mechanical fatigue from hydration cycling; 3) long-term strain drift associated with irreversible structural aging. Additionally, through special and designed engineering and installation strategy, **flow** and **pressure** can be measured as well [patent as reference?]. These measurements are fundamental to SoO monitoring and provide indirect but valuale information for SoH analysis.

	The OFMS (FBGs) for in-situ measurement provides special advantages. First, spatially resolved sensing across a cell and stack by multiplexing and enable >10 FBGs in a single fiber. Second, the OFMS provides high resolution and ≤ 0.1°C temperature resolution and ≤ 1 με strain resolution are possible. Third, the OFMS offers high data quality and has long-term stability with minimal drift over thousands of operating hours. This enables continuous time logged data acquisition that are critical for SoH analysis and diagnostics.
R&D Plan	The FBGs sense multiple variables at the same time, such as temperature and strain and this is known as cross-sensitivity. To get accurate sensing and measurement, the cross-sensitivity needs to be treated and decoupled. This can be done through either multipoint measurement scheme or special modified fiber. TNO has experience with both. Depends on the application and use case, the right engineering strategy can be applied to achieve low or none cross-sensitivity. This is critical for FCs in dynamic operation use cases such as automotive, as those applications involve fast transients, where SoO effects dominate measured signals and mask aging. Data quality can be further improved by AI-assisted modelling and analysis. AI and machine learning (ML) algorithms will use full spectral responses instead of just peak positions, surpassing the ~1 pm resolution limit of conventional FBG intererogation methods. This allows accurate discrimination of temperature (**0.1 °C**) and strain (**0.5 με**) even in noisy or dynamic settings. The result will be a reusable geometry–performance library and a versatile spectral analysis toolbox. Validation will begin with static calibration under controlled thermal, pressure, and strain conditions to determine sensitivity, linearity, repeatability and cross sensitivity, and generate spectral datasets for AI/ML model training. Dynamic tests will follow, correlating spectral responses with parameters like vibration frequency, and flow rate. Environmental durability will be assessed by exposing coated fibres and assemblies to corrosive and humid atmospheres to verify long-term stability. System-level validation in FCs will test integrated arrays in realistic electrochemical environments at user site. The location of integration is use-case specific and will be investigated together with partners for the relevant FC design and systems. The sensors used in FC should secure the robustness of the integration. A further objective is to achieve high quality embedding fibers inside stacks. The embedded fibers should not cause flow disturbance which can affect the FC operation. The fibers remain mechanical reliability against long-term fatigue. The fiber installation interface and sealing ensure compression uniformity. The work plan is described in the following work packages and tasks. <u>WP1. Use case requirements and specifications [M1-M2]</u> T1.1 identify the fuel cell system and stack for SoO and SoH study. FC stack and relevant technical specifications and information provided by SME. T1.2 FC stack operational parameters linked to the long-term durability of key components, such as PEM membrane, will be identified by SEM. TNO will identify sensors with proven capability to capture these parameters within the stack. SoO monitoring sensing requirements will be developed by TNO and SME partner(s). *D1*. Fuel cell SoO monitoring and SoH analysis sensing requirement report. <u>WP2. OFMS sensing system design and development [M3-M6]</u> T2.1 Multiparameter sensor array design. Based on the selected FC stack, design the sensor array system including the installation strategy and necessary engineering solutions for the robustness and the installation. Propose the first sealing and fiber feedthrough design. T2.2 Optical components selection, procurement and engineering. T2.3 Sensing parameters and targets testing and validation. *D2*. Optical components with validated sensing performance and the engineering solutions <u>WP3. Sensor system prototype and calibration [M7-M10]</u> T3.1 Sensor component coating and microstructure development and manufacturing by TNO. T3.2 Multiparameter sensor cross-sensitivity cancellation engineering and data processing. T3.3 OFMS system calibration for sensing specifications.

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	D3. Sensor system engineering model ready for employment. *M1*. Sensor system lab demonstration WP4. First Integration of OFMS in FC [M10-M13] T4.1 Integration of optical fiber strategy and engineering procedure?, methods for designed FC stack. TNO and SME partner will work together to develop suitable integration plan. T4.2 Engineering practice of FC component for embedding optical fibers. T4.3 Engineering and integration of optical fiber in FC components. Test the sensor functionality and validation and calibration. *D4.* FC stack with installed OFMS is ready for testing. WP5. Testing and monitoring data acquisition [M14-M17] T5.1 Testing protocol and plan will be made for the purpose of SoO validation and SoH analysis. The testing program should include scenario: 1) normal operation; 2) dynamic operation; 3) stressed operation; 4) operation with (controlled) damaged components (e.g. membrane); 5) any other important and relevant operations defined by user. T5.2 Acquisition of time logged OFMS data with other sensors in the multimodal system during operation and running of FC testing program. T5.2 Multimodal sensor data fusion with high accurate time synchronisation. *D5*. Multimodal sensing data including OFMS data sequences are structured. Also a first model??? WP6. Information distillation for SoO monitoring and SoH analysis [M18-M20] T6.1 Data processing for key operation parameters such as, T-distribution, strain, flow. AI/ML algorithms will be applied for improved data quality in term of higher S/N. T6.2 The SoH analysis will be carried out based on controlled damage testing from WP5. ML algorithms will be applied for efficient information distillation to identify degradation with operation parameters. *D6*. Sensing result analysis report. *M2*. OFMS for SoO and SoH demonstration. WP7. OFMS system packaging and final installation [M21-M24] T7.1 Improved installation strategy and plan T7.2 Sensor packaging design including industrial prototype of sealing and interfacing. T7.3 Install and embed OFMS in industrial FC stack T7.4 Testing and acquisition SoO monitoring sensing data. T7.5 Data processing and final validation. *D7*. FC stack with embedded OFMS system. *D8*. Final project result and report. *M3*. Completion of OFMS in industrial FC stack.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO as the independent R&D research institute is responsible for • Defining and maturing the optical fiber sensing concept into a deployable monitoring/diagnostic subsystem for PEM fuel cell stacks. • Bridging fundamental sensing science and engineering knowledge, reducing technical risk for SMEs by delivering validated sensing methods, integration guidelines, and diagnostic performance evidence. • Ensuring sensor development, and validation and implementation using test protocols agreed with Korean partners. TNO holds key IPs on OFMS system including: 3 patents, dedicated tools and engineering methods, most advanced research and testing labs and facilities and more than 40 years Know-Hows in Fiber Optics domain.
Expected R&R of Korean SME Partner	SME1: Sensor product development company or fuel cell system monitoring and control engineering company SME2: Fuel cell OEM or end-user – provide testing fuel cell stack and fuel cell testing facility and provide user specification and requirement SMEs are expected to bring knowledge and working on: • Stack hardware and BoP integration • Operation profiles, use cases, and market requirements • Field trials, application validation, and productization pathways • System controller integration and commercialization

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project delivers a validated optical fiber–based monitoring solution that improves internal observability of PEM fuel-cell operation, reduces uncertainty in health diagnostics, accelerates industrial adoption, and strengthens the competitiveness of Korean fuel-cell SMEs within the global hydrogen economy. Technical achievements: 1. Validated OFMS with demonstration of spatially resolved multiparameter SoO monitoring inside or at critical interface of fuel cell stacks. This will reduce internal state observability gaps that limit conventional fuel-cell diagnostics. 2. Improved SoO observability and diagnostic inputs will supports earlier dection of abnormal operation and incipient failure mechanisms. 3. Integration and design guidelines for OEMs and SMEs will lower technical barriers for adoption by fuel cell manufactures and industry. Industrial outcomes: 1. Reduced development and validation risks for SMEs shortens time-to-market and lowers integration costs. 2. Enabling advanced monitoring-enabled fuel cell products. This will strengthen global competitiveness of participating SMEs. Market impacts: 1. Improved fuel cell reliability and lifetime will contribute to lower total cost of ownership (TCO). 2. Contribution to hydrogen ecosystem maturity. This unique collaboration enables cross-reginal technology transfer through structured knowledge exchange between TNO and Korean SMEs, combining European sensor R&D excellence with Korean fuel-cell manufacturing and market expertise. By development of a shared technical language for SoO/SoH diagnostics, we lay foundation for future innovations. It positions the consortium for follow-on projects, standardization efforts, and global leadership in diagnostic-enabled hydrogen technologies.

6. References

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